

Experiment 6 – MC

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COURSE CODE:MLA0305

COURSE NAME:REINFORCEMENT LEARNING

SLOT: B

6.Monte Carlo Prediction andControl for Reinforcement Learning

Aim:

To implement Monte Carlo Prediction and Control methods in Reinforcement Learning to estimate the value function and determine the optimal policy through repeated episodes.

Procedure:

1. Define the states, actions, and rewards of the environment.
2. Initialize the value function and action-value function .
3. Generate episodes by selecting actions and observing the resulting rewards.
4. Calculate the return from each visited state-action pair.
5. Update the value function using the average returns obtained from multiple episodes.
6. For Monte Carlo Prediction, estimate the value of each state from the observed returns.
7. For Monte Carlo Control, update the action-value function and improve the policy by selecting the action with the highest value.
8. Repeat the process for several episodes until the values and policy converge.
9. Display the estimated values and the resulting optimal policy.

R-CODE:

MONTE CARLO METHOD - REAL TIME EXAMPLE

Example: Daily Coffee Shop Profit Estimation

One Complete R Program

Clear Console

```
cat("\014")
```

```
set.seed(123)
```

REAL-TIME DATA

```
days <- paste("Day",1:10)
```

```
daily_profit <- sample(seq(450,900,50),10,replace=TRUE)
```

PROCESS 1 : EPISODE

```
cat("=====\n")
```

```
cat("PROCESS 1 : EPISODE\n")
```

```
cat("=====\n\n")
```

```
cat("Explanation:\n")
```

```
cat("Each episode represents one working day in a coffee shop.\n")
```

```
cat("Every day has a different profit depending on customer visits.\n\n")
```

```
episode_colors <- rainbow(length(daily_profit))
```

```
bp <- barplot(daily_profit,
```

```
    names.arg=days,
```

```
    col=episode_colors,
```

```
    border="black",
```

```
    main="Episode: Daily Coffee Shop Profit",
```

```

        xlab="Working Days",

        ylab="Profit (Rs)",

        ylim=c(0,max(daily_profit)+150))

text(bp,daily_profit,

     labels=daily_profit,

     pos=3,

     cex=0.8)

readline("Press ENTER to Continue...")

# PROCESS 2 : COLLECT REWARDS

cat("\014")

cat("=====\n")

cat("PROCESS 2 : COLLECT REWARDS\n")

cat("=====\n\n")

cat("Explanation:\n")

cat("The profit earned each day becomes the reward collected\n")

cat("by the Monte Carlo agent.\n\n")

reward <- daily_profit

reward_colors <- heat.colors(length(reward))

bp <- barplot(reward,

              names.arg=days,

              col=reward_colors,

              border="black",

              main="Collect Rewards (Daily Profit)",

              xlab="Working Days",

```

```

        ylab="Reward (Profit)",

        ylim=c(0,max(reward)+150))

text(bp,reward,

     labels=reward,

     pos=3,

     cex=0.8)

readline("Press ENTER to Continue...")

# PROCESS 3 : COMPUTE RETURN

cat("\014")

cat("=====\n")

cat("PROCESS 3 : COMPUTE RETURN\n")

cat("=====\n\n")

cat("Explanation:\n")

cat("Monte Carlo computes the return.\n")

cat("Here the return is the cumulative profit earned up to\n")

cat("each working day.\n\n")

returns <- cumsum(reward)

return_colors <- terrain.colors(length(returns))

bp <- barplot(returns,

              names.arg=days,

              col=return_colors,

              border="black",

              main="Compute Return (Cumulative Profit)",

              xlab="Working Days",

```

```

        ylab="Cumulative Profit",

        ylim=c(0,max(returns)+300))

text(bp,returns,

     labels=returns,

     pos=3,

     cex=0.8)

readline("Press ENTER to Continue...")

# PROCESS 4 : UPDATE VALUES

cat("\014")

cat("=====\n")

cat("PROCESS 4 : UPDATE VALUES\n")

cat("=====\n\n")

cat("Explanation:\n")

cat("Monte Carlo updates the estimated value.\n")

cat("Here we estimate the average daily profit after each day.\n\n")

value_estimate <- cumsum(reward)/(1:length(reward))

value_colors <- topo.colors(length(value_estimate))

bp <- barplot(value_estimate,

              names.arg=days,

              col=value_colors,

              border="black",

              main="Update Values (Average Profit Estimate)",

              xlab="Working Days",

              ylab="Estimated Average Profit",

```

```

        ylim=c(0,max(value_estimate)+150))

text(bp,value_estimate,

     labels=round(value_estimate,1),

     pos=3,

     cex=0.8)

readline("Press ENTER to Continue...")

# PROCESS 5 : REPEAT

cat("\014")

cat("=====\n")

cat("PROCESS 5 : REPEAT\n")

cat("=====\n\n")

cat("Explanation:\n")

cat("Monte Carlo repeats the same process for many episodes.\n")

cat("As more working days are observed, the estimated average\n")

cat("profit becomes more stable.\n\n")

extra_profit <- sample(seq(450,900,50),20,replace=TRUE)

repeat_estimate <- cumsum(extra_profit)/(1:length(extra_profit))

# Different color for every episode

repeat_colors <- c(

  "red","orange","gold","yellow","yellowgreen",

  "green","springgreen","turquoise","cyan","deepskyblue",

  "dodgerblue","blue","slateblue","purple","violet",

  "magenta","hotpink","brown","coral","darkred"

)

```

```

bp <- barplot(repeat_estimate,

              col=repeat_colors,

              border="black",

              main="Repeat Process (Improving Average Profit Estimate)",

              xlab="Episode (Working Day)",

              ylab="Estimated Average Profit",

              ylim=c(0,max(repeat_estimate)+150))

text(bp,

      repeat_estimate,

      labels=round(repeat_estimate,1),

      pos=3,

      cex=0.7)

readline("Press ENTER for Final Summary...")

# FINAL SUMMARY

cat("\014")

cat("=====\n")

cat("FINAL SUMMARY OF MONTE CARLO METHOD\n")

cat("=====\n\n")

cat("Episode      : One working day in the coffee shop.\n")

cat("Collect Reward : Profit earned each day.\n")

cat("Compute Return : Cumulative profit over days.\n")

cat("Update Values  : Running average daily profit.\n")

cat("Repeat        : Observe more days to improve estimate.\n\n")

cat("Daily Profit:\n")

```

```
print(daily_profit)
```

```
cat("\nAverage Daily Profit = ",round(mean(daily_profit),2),"Rs\n")
```

```
cat("\nEstimated Profit after all episodes = ",
```

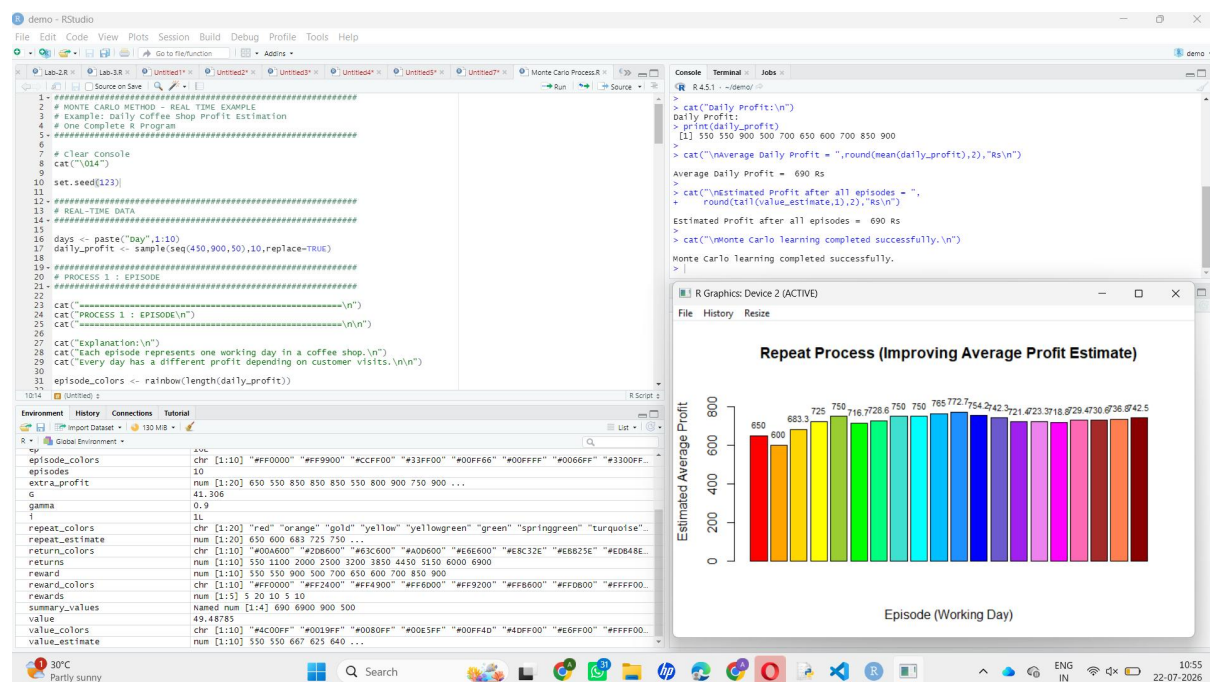
```
round(tail(value_estimate,1),2),"Rs\n")
```

```
cat("\nMonte Carlo learning completed successfully.\n")
```

Result:

The **Monte Carlo Prediction and Control** algorithm was successfully implemented using **RStudio**. The program estimated the value function from sampled episodes and improved the policy based on the accumulated returns. The final output displayed the **state values, action values, optimal policy, and learning performance graph**. Thus, the optimal policy was successfully obtained using Monte Carlo methods in Reinforcement Learning.

Full screenshot of Rstudio:



Summary Result:

```
demo - RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
R 4.5.1 - demo
> cat("-----\n")
> cat("FINAL SUMMARY OF MONTE CARLO METHOD\n")
FINAL SUMMARY OF MONTE CARLO METHOD
> cat("-----\n")

>
> cat("Episode : one working day in the coffee shop.\n")
Episode : one working day in the coffee shop.
> cat("Collect Reward : Profit earned each day.\n")
Collect Reward : Profit earned each day.
> cat("Compute Return : Cumulative profit over days.\n")
Compute Return : Cumulative profit over days.
> cat("Update values : Running average daily profit.\n")
Update values : Running average daily profit.
> cat("Repeat : observe more days to improve estimate.\n")
Repeat : observe more days to improve estimate.

>
> cat("Daily Profit:\n")
Daily Profit:
> print(daily_profit)
[1] 550 550 900 900 700 650 600 700 850 900
> cat("\nAverage Daily Profit = ",round(mean(daily_profit),2),"Rs\n")
Average Daily Profit = 690 Rs
> cat("\nEstimated Profit after all episodes = ",
+ round(tail(value_estimate,1),2),"Rs\n")
Estimated Profit after all episodes = 690 Rs
> cat("\nMonte Carlo learning completed successfully.\n")
Monte Carlo learning completed successfully.
> |
```

Visualization Result:

